Perceptron

CS60050: Machine Learning

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IIT Kharagpur

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Agenda

- § Introducing the concept artificial neurons
- § Understand the perceptron learning algorithm and proving its convergence

Resources

- § Artificial Intelligence: A Modern Approach by Stuart Russell and Peter Norvig.
- § NPTEL Lectures on Neural Networks and Applications [Lecture 12 & 16] by Prof. Somnath Sengupta, IIT Kharagpur.

Biological Neural Network

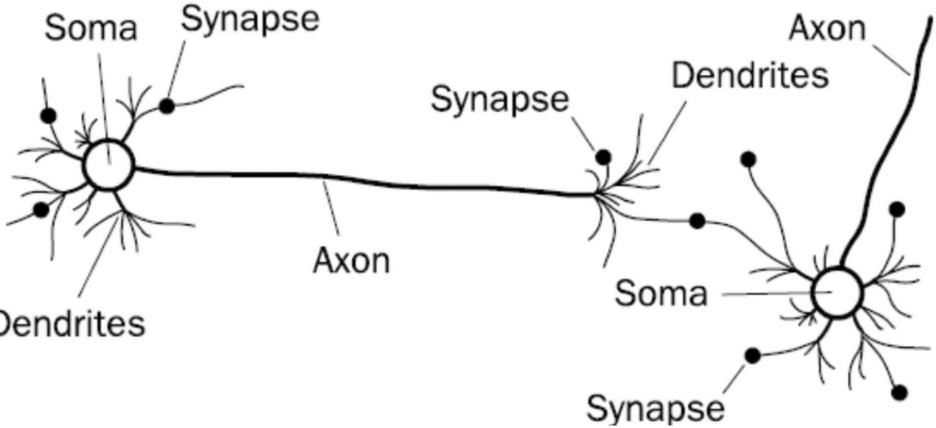
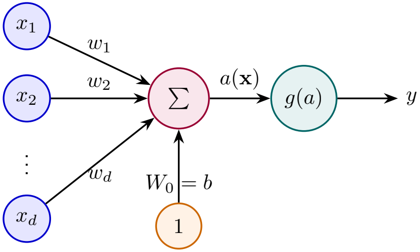


Fig. credit: *'Prediction of crude oil viscosity using feed-forward back-propagation neural network (FFBPNN)', Petroleum & Coal 2012.*

An Interesting CVPR 2017 Talk

Artificial Neuron



$$\boldsymbol{w} = [w_1 \ w_2 \ \ldots \ w_d]^T \text{ and } \boldsymbol{x} = [x_1 \ x_2 \ \ldots \ x_d]^T$$

$$a(\boldsymbol{x}) = b + \sum_{i=1}^{d} w_i x_i = [\boldsymbol{w}^T \quad b] \begin{bmatrix} \boldsymbol{x} \\ 1 \end{bmatrix} = \boldsymbol{w}^T \boldsymbol{x} + b$$

$$y = g(a(\boldsymbol{x}))$$

Terminologies

- $\boldsymbol{x}$: input, $\boldsymbol{w}$: weights, b: bias
- a: pre-activation (input activation)
- g: activation function
- y: activation (output activation)

Perceptron

The New York Times

Electronic 'Brain' Teaches Itself

JULY 13, 1958

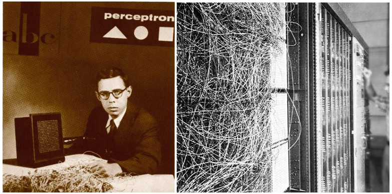
The Navy last week demonstrated the embryo of an electronic computer named the Perceptron which, when completed in about a year, is expected to be the first non-living mechanism able to "perceive, recognize and identify its surroundings without human training or control." Navy officers demonstrating a preliminary form of the device in Washington said they hesitated to call it a machine because it is so much like a "human being without life."

Dr. Frank Rosenblatt, research psychologist at the Cornell Aeronautical Laboratory, Inc., Buffalo, N. Y., designer of the Perceptron, conducted the demonstration. The machine, he said, would be the first electronic device to think as the human brain. Like humans, Perceptron will make mistakes at first, "but it will grow wiser as it gains experience," he said.

recognize the difference between right and left, almost the way a child learns.

When fully developed, the Perceptron will be designed to remember images and information it has perceived itself, whereas ordinary computers remember only what is fed into them on punch cards or magnetic tape.

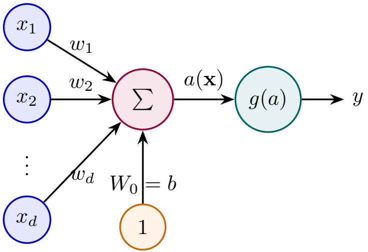
Later Perceptrons, Dr. Rosenblatt said, will be able to recognize people and call out their names. Printed pages, longhand letters and even speech commands are within its reach. Only one more step of development, a difficult step, he said, is needed for the device to hear speech in one language and instantly translate it to speech or writing in another language.



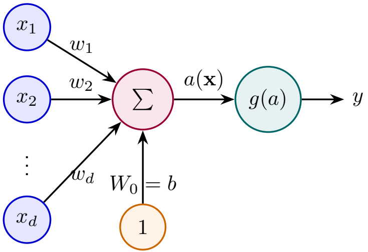
Perceptron

$\boldsymbol{x} \in \mathbb{R}^d$ and $y \in \{0, 1\}$ – Binary Classification

$$g(a) = \begin{cases} 1, & a \geq 0 \\ 0, & a < 0 \end{cases}$$ (Rosenblatt, 1957)



Perceptron



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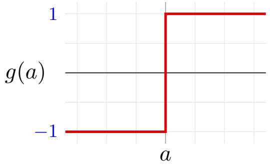
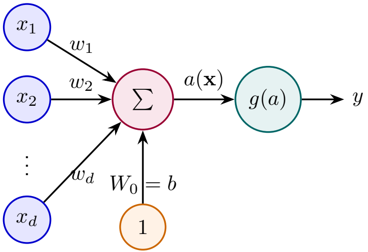


Perceptron



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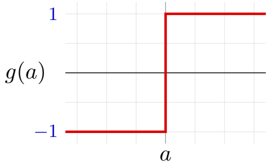

The perceptron classification rule, thus, translates to

$$y = \begin{cases} 1, & \boldsymbol{w}^T\boldsymbol{x} + b \geq 0 \\ -1, & \boldsymbol{w}^T\boldsymbol{x} + b < 0 \end{cases}$$

Remember
$\boldsymbol{w}^T\boldsymbol{x} + b = 0$ represents a hyperplane.

Perceptron Learning Algorithm

Given a training set:
$\{\boldsymbol{x}^{(i)}, y^{(i)} : \boldsymbol{x}^{(i)} \in \mathbb{R}^d, y^{(i)} \in \{-1, 1\}\}$ for $i = 1, 2, \dots, N$

1. Start with $k = 1$ and $\boldsymbol{w}^{(k)} = [0\ 0\ \dots\ 0]^T$
2. Loop until all examples are correctly classified:

   $\boldsymbol{w}^{(k+1)} \leftarrow \boldsymbol{w}^{(k)} + y^{(k)}\boldsymbol{x}^{(k)}$ if $\boldsymbol{x}^{(k)}$ is misclassified.
   [This means $\boldsymbol{w}^{(k+1)} \leftarrow \boldsymbol{w}^{(k)} + \boldsymbol{x}^{(k)}$ or $\boldsymbol{w}^{(k+1)} \leftarrow \boldsymbol{w}^{(k)} - \boldsymbol{x}^{(k)}$]

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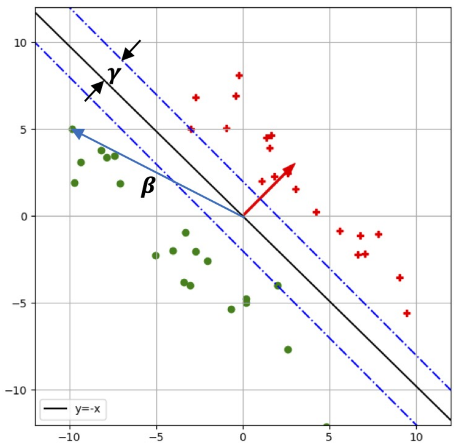
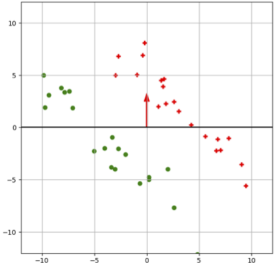


Convergence Theorem:-

For a finite and linearly separable set of data, the perceptron learning algorithm will find a linear separator in at most $\frac{\beta^2}{\gamma^2}$ iterations where the maximum length of any data point is β and γ is the maximum margin of the linear separators.

Perceptron Learning Algorithm



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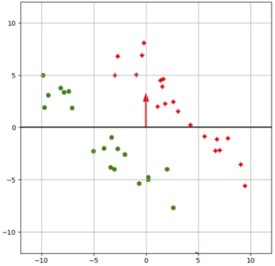
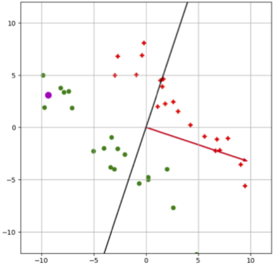
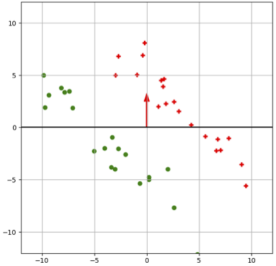
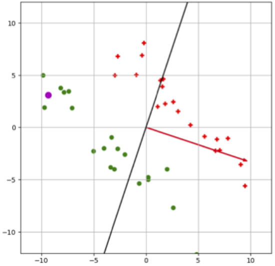
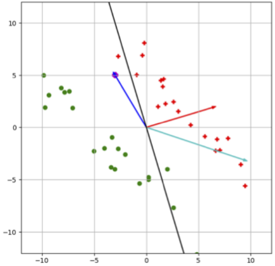




Perceptron Learning Algorithm







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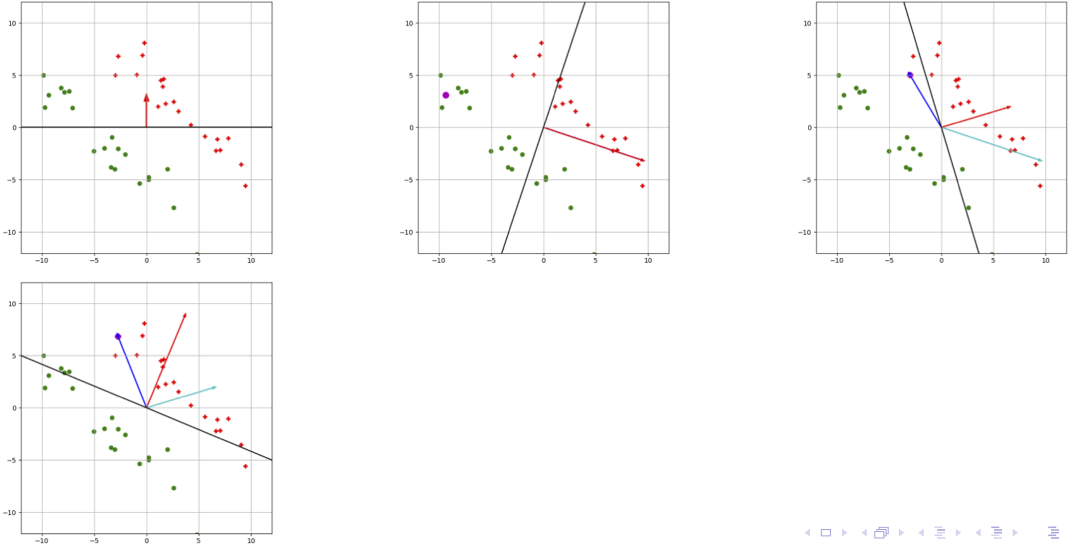

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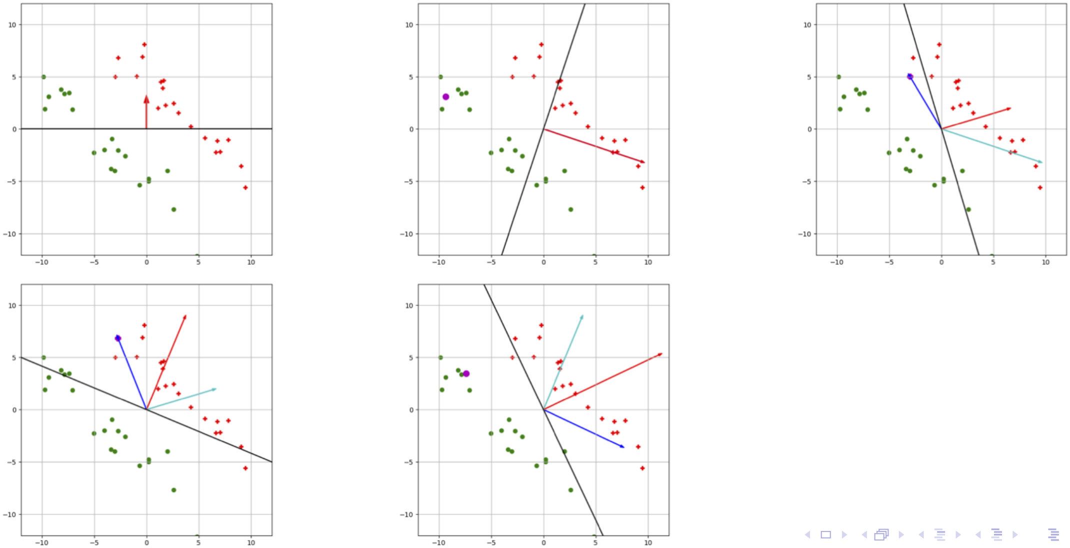

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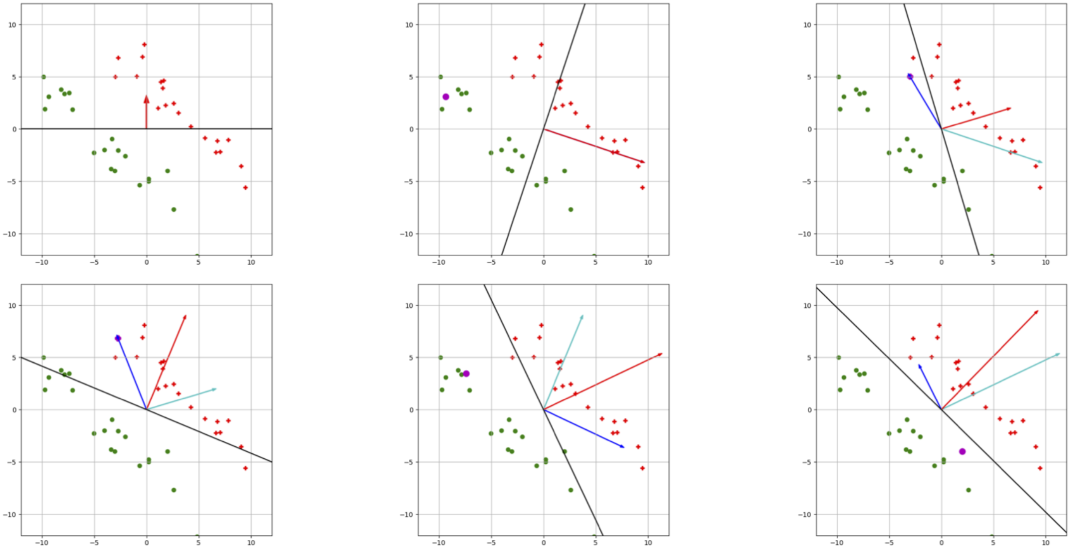
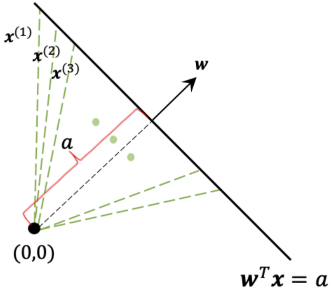


Perceptron Learning Algorithm – Convergence Proof

Some Important Brush-ups:-



For a hyperplane $\boldsymbol{w}^T\boldsymbol{x} = a$, $\boldsymbol{w}$ is the normal vector (often assumed to be unit normal vector) to the hyperplane and a is the distance from the origin.

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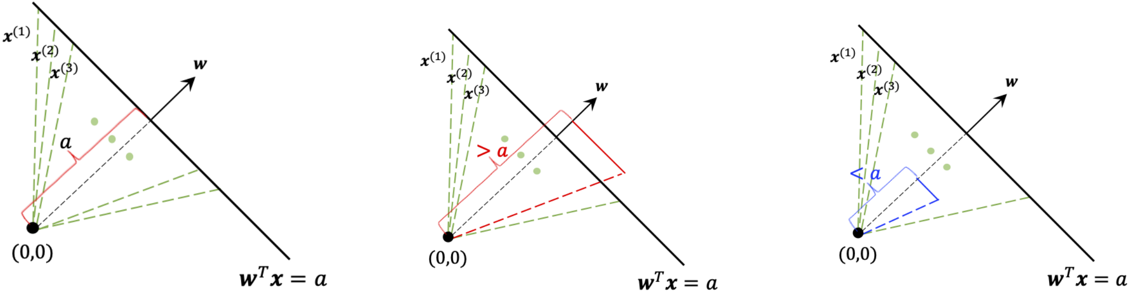

For a hyperplane $\boldsymbol{w}^T\boldsymbol{x} = a$, $\boldsymbol{w}$ is the normal vector (often assumed to be unit normal vector) to the hyperplane and a is the distance from the origin.

Thus a hyperplane with normal vector $\boldsymbol{w}$, can separate two sets of points by examining if $\boldsymbol{w}^T\boldsymbol{x} - a$ is > 0 or < 0. (Remember our notation says $-a$ is nothing but b).

Perceptron Learning Algorithm – Convergence Proof

- Thus a simple classification rule can be $\boldsymbol{w}^T\boldsymbol{x} + b > 0$ or < 0.
- For simplicity, let us assume that the hyperplane passes through the origin. This does not affect the problem in our hand as it just shifts the origin. But that means $b = 0$. Thus, the classification rule becomes:

$$\operatorname{sgn}\left(\boldsymbol{w}^T\boldsymbol{x}^{(i)}\right) = y^{(i)} \quad \forall(\boldsymbol{x}^{(i)}, y^{(i)})$$

- There may not exist such a hyperplane that can separate two sets of data. If it exists, then the data is known to be **linearly separable**.
- A more formal definition is - Two sets P and N of d-dimensional points are said to be **linearly separable** if there exist $d+1$ real numbers $\boldsymbol{w} = (w_0, w_1, \ldots, w_d)$ such that every point $\boldsymbol{x} = (1, x_1, x_2, \ldots, x_d) \in P$ satisfies $\operatorname{sgn}\left(\boldsymbol{w}^T\boldsymbol{x}\right) = 1$ and $\boldsymbol{x} = (1, x_1, x_2, \ldots, x_d) \in N$ satisfies $\operatorname{sgn}\left(\boldsymbol{w}^T\boldsymbol{x}\right) = -1$.

Perceptron Learning Algorithm – Convergence Proof

Without loss of generalization we will also assume the following while proving the convergence theorem.

- We normalize all the points $\boldsymbol{x}^{(i)}$ so that $max\ \|\boldsymbol{x}^{(i)}\| = 1$, *i.e.*, maximum length of any data point is 1. Notice that this does not affect the algorithm or the solution as, $sgn(\boldsymbol{w}^T\boldsymbol{x}^{(i)}) = sgn\left(\boldsymbol{w}^T\frac{\boldsymbol{x}^{(i)}}{\|\boldsymbol{x}^{(i)}\|}\right)$.
- **Margin of separation:** This is the minimum distance between any data point and the hyperplane $\boldsymbol{w}^T\boldsymbol{x} = 0$. Among all the hyperplanes, a hyperplane having the maximum margin of separation is called the *maximum margin separator*.

$$\gamma = \min_{\boldsymbol{x}} |\boldsymbol{w}^T\boldsymbol{x}|$$

Perceptron Learning Algorithm – Convergence Proof

§ Let $\boldsymbol{w}^*$ be the normal to the maximum margin linear separator. We want our $\boldsymbol{w}$ in the perceptron algorithm in each step getting closer and closer to this (sort of) ideal $\boldsymbol{w}^*$.

§ This means that the inner product between $\boldsymbol{w}$ and $\boldsymbol{w}^*$ should increase with each update.

§ Remember the update rule: $\boldsymbol{w}' = \boldsymbol{w} + \boldsymbol{x}^{(i)} y^{(i)}$

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  So, $(\boldsymbol{w}')^T \boldsymbol{w}^* = (\boldsymbol{w} + \boldsymbol{x}^{(i)})^T \boldsymbol{w}^*$ (assuming $y^{(i)} = +1$)

$$= \underbrace{\boldsymbol{w}^T \boldsymbol{w}^*}_{\boxed{\text{Previous } \boldsymbol{w}^T \boldsymbol{w}^*}} + \underbrace{(\boldsymbol{x}^{(i)})^T \boldsymbol{w}^*}_{\boxed{\text{This is at least } \gamma}}$$

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§ Similarly for a negative training example, $(\boldsymbol{w}')^T \boldsymbol{w}^* = (\boldsymbol{w} - \boldsymbol{x}^{(i)})^T \boldsymbol{w}^* = \boldsymbol{w}^T \boldsymbol{w}^* - (\boldsymbol{x}^{(i)})^T \boldsymbol{w}^*$

$(\boldsymbol{x}^{(i)})^T \boldsymbol{w}^*$ is negative as the example is negative and thus $-(\boldsymbol{x}^{(i)})^T \boldsymbol{w}^* \geq \gamma$

§ So, in both cases $(\boldsymbol{w}')^T \boldsymbol{w}^*$ is increasing by at least γ. Which means length of $\boldsymbol{w}'$ increases with each update $\rightarrow$ **Observation 1**

Perceptron Learning Algorithm – Convergence Proof

- length of $\boldsymbol{w}'$ is given by $\|\boldsymbol{w}'\|^2$. Let us see, how this changes with each update.
- $\|\boldsymbol{w}'\|^2 = \boldsymbol{w}'^T\boldsymbol{w}' = (\boldsymbol{w} + \boldsymbol{x}^{(i)}y^{(i)})^T(\boldsymbol{w} + \boldsymbol{x}^{(i)}y^{(i)}) = \|\boldsymbol{w}\|^2 + 2y^{(i)}(\boldsymbol{x}^{(i)})^T\boldsymbol{w} + (y^{(i)})^2\|\boldsymbol{x}^{(i)}\|^2$
- Let us consider the case $y^{(i)} = +1$ first.

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  $\|\boldsymbol{w}'\|^2 = \|\boldsymbol{w}\|^2 + 2(\boldsymbol{x}^{(i)})^T\boldsymbol{w} + \|\boldsymbol{x}^{(i)}\|^2$

  This is negative as update occurs only on misclassification and for $y^{(i)} = +1$, misclassification means $(\boldsymbol{x}^{(i)})^T\boldsymbol{w} \leq 0$.

  This is ≤ 1, as we assumed every point is rescaled to unit ball.

- So, $\|\boldsymbol{w}'\|^2 \leq \|\boldsymbol{w}\|^2 + 1$

Perceptron Learning Algorithm – Convergence Proof

- Let us now consider the case $y_i = -1$.
- $\|\boldsymbol{w}'\|^2 = \|\boldsymbol{w}\|^2 - 2\boldsymbol{x}_i^T\boldsymbol{w} + \|\boldsymbol{x}_i\|^2$

Perceptron Learning Algorithm – Convergence Proof

- Let us now consider the case $y_i = -1$.
- $\|\boldsymbol{w}'\|^2 = \|\boldsymbol{w}\|^2 - 2\boldsymbol{x}_i^T\boldsymbol{w} + \|\boldsymbol{x}_i\|^2$
- Using similar arguments, $\boldsymbol{x}_i^T\boldsymbol{w} \geq 0$, so, $-2\boldsymbol{x}_i^T\boldsymbol{w} \leq 0$ and also $\|\boldsymbol{x}_i\|^2 \leq 1$
- So, here also $\|\boldsymbol{w}'\|^2 \leq \|\boldsymbol{w}\|^2 + 1$

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- So, the length of the updated vector can get modified and that too in a controlled way. $\rightarrow$ **Observation 2**

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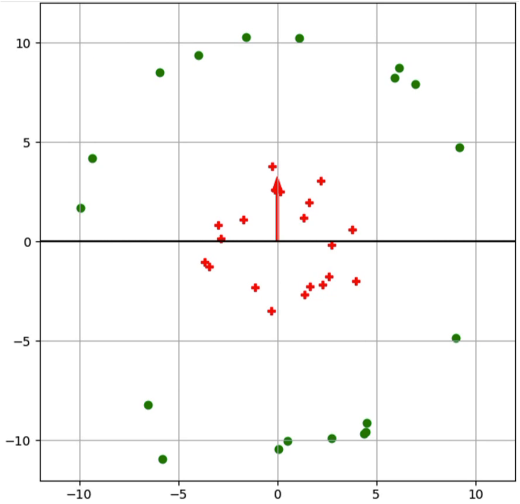
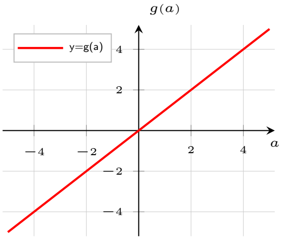
- So, the length of the updated vector can get modified and that too in a controlled way. $\rightarrow$ **Observation 2**
- Say after k iterations, $\boldsymbol{w}$ is denoted as $\boldsymbol{w}_k$.
- So observation 1 implies, $\boldsymbol{w}_k^T\boldsymbol{w}^* \geq k\gamma$ and observation 2 implies $\|\boldsymbol{w}_k\|^2 \leq k$ [remember the initial value of $\boldsymbol{w}$ is 0]
- Now we have to apply the Cauchy-Swartz inequality.
- $\|\boldsymbol{w}_k\|\|\boldsymbol{w}^*\| \geq \boldsymbol{w}_k^T\boldsymbol{w}^*$, but $\|\boldsymbol{w}^*\| = 1$, so, $\|\boldsymbol{w}_k\| \geq \boldsymbol{w}_k^T\boldsymbol{w}^* \geq k\gamma$ [observation 1]
- This implies $k^2\gamma^2 \leq \|\boldsymbol{w}_k\|^2 \leq k$ [observation 2]
- So, $k \leq \frac{1}{\gamma^2}$

Animation Demo

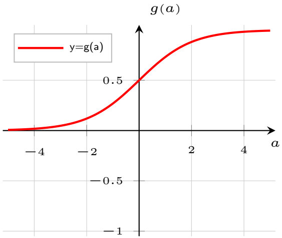


Popular Choices of Activation Functions



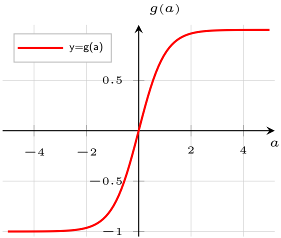
Linear activation function

- $g(a) = a$
- Unbounded
- $g'(a) = 1$



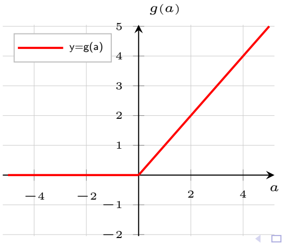
Sigmoid activation function

- $g(a) = \sigma(a) = \frac{1}{1+\exp(-a)}$
- Bounded $(0, 1)$
- Always positive
- $g'(a) = g(a)(1 - g(a))$



tanh activation function

- $g(a) = \tanh(a) = \frac{\exp(a)-\exp(-a)}{\exp(a)+\exp(-a)}$
- Bounded $(-1, 1)$
- Can be positive or negative
- $g'(a) = 1 - g^2(a)$



ReLU activation function

- $g(a) = \max(0, a)$
- Bounded below by 0
- But not upper-bounded
- $g'(a) = \begin{cases} 1, & a \geq 0 \\ 0, & a < 0 \end{cases}$